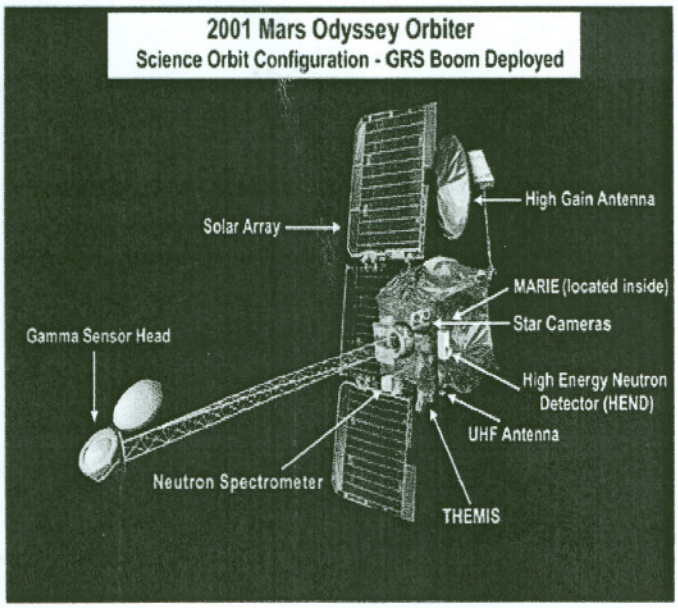
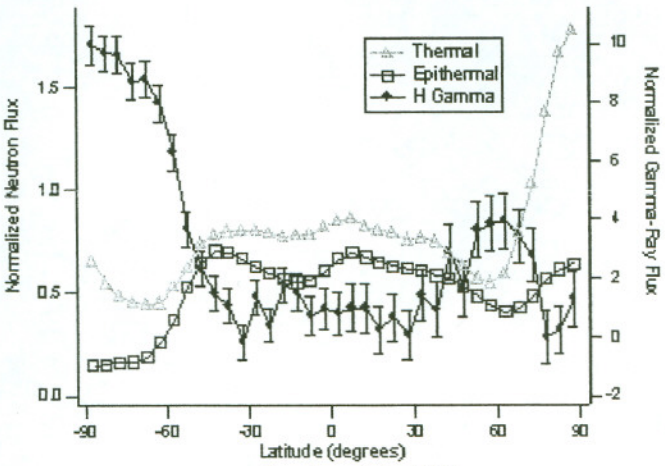
ANALTICAL CUMULATIVE EXAMINATION
5TH December 2003

The Odyssey spacecraft currently in orbit around Mars, has three instruments for analytical geochemistry, a gamma ray spectrometer, a high energy neutron detector and a neutron spectrometer (Fig. 1). An example of the first data obtained by these instruments is shown in Fig. 2 and the relevant portion of the Chart of the Nuclides is shown in Fig. 3.

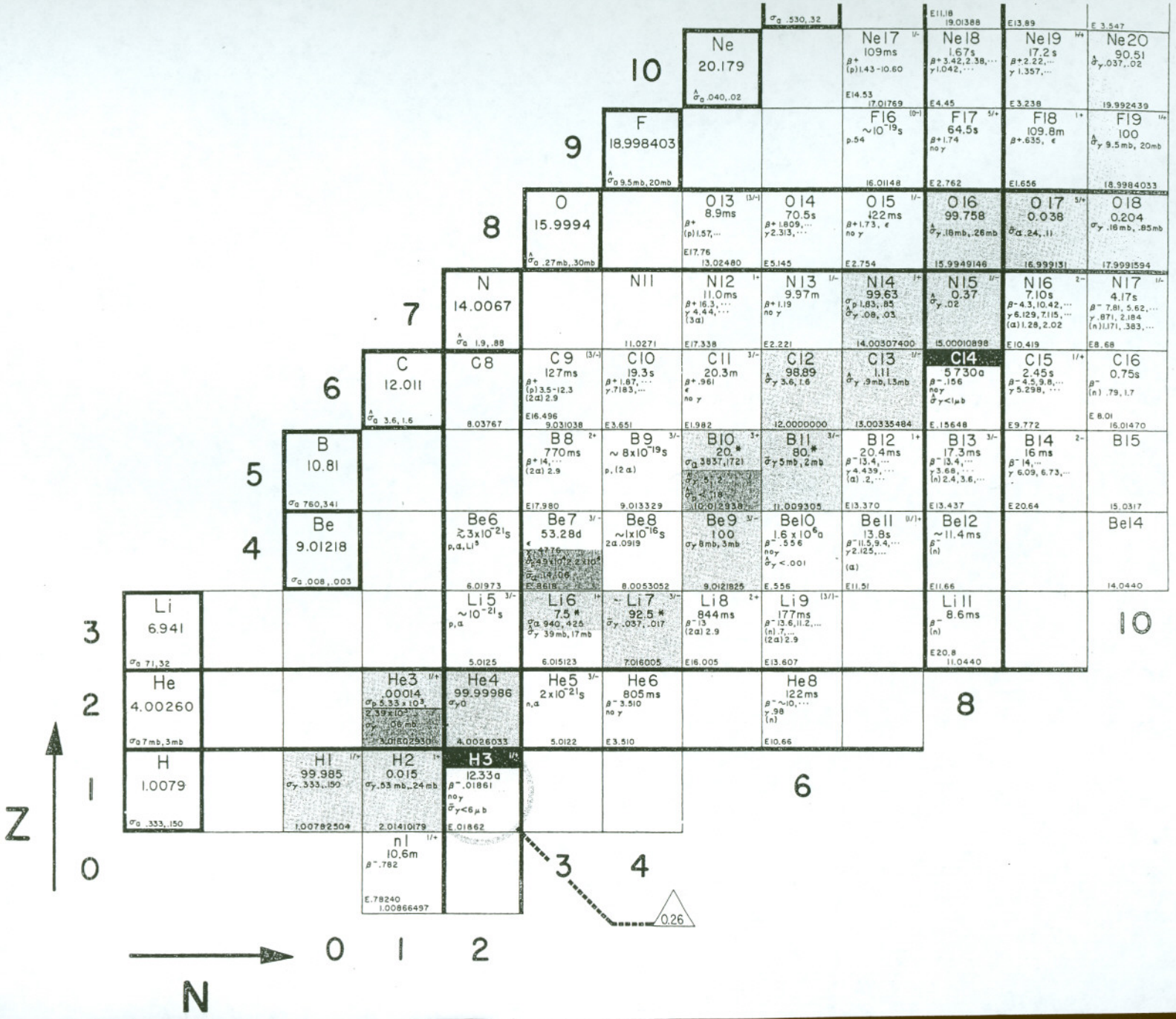
(a) Explain how these three instruments work. Credit will be given for a knowledge of the basic underlying nuclear chemistry, to the use of quantitative explanations, for sensitivity to the challenges of obtaining quantitative information from these techniques, and to how these methods are integrated into the spacecraft system. Credit will also be given for intelligent guesses.

(b) Describe the interpretation of the data in Fig. 2 offered by the science teams responsible for these three instruments. To what extent do you agree with these interpretations. What are the assumptions and uncertainties inherent in their arguments. What measurements would you make to test their ideas.





Boynton et al., 2002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H2 0.015
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